# Forest Queries

**Time limit:** 1.00 s **Memory limit:** 512 MB

You are given an $n \times n$ grid representing the map of a forest. Each square is either empty or contains a tree. The upper-left square has coordinates $(1, 1)$, and the lower-right square has coordinates $(n, n)$.

Your task is to process $q$ queries of the form: how many trees are inside a given rectangle in the forest?

## Input

The first input line has two integers $n$ and $q$: the size of the forest and the number of queries.

Then, there are $n$ lines describing the forest. Each line has $n$ characters: `.` is an empty square and `*` is a tree.

Finally, there are $q$ lines describing the queries. Each line has four integers $y_1, x_1, y_2, x_2$ corresponding to the corners of a rectangle.

## Output

Print the number of trees inside each rectangle.

## Constraints

- $1 \le n \le 1000$
- $1 \le q \le 2 \cdot 10^5$
- $1 \le y_1 \le y_2 \le n$
- $1 \le x_1 \le x_2 \le n$

## Example

| Input | Output |
|---|---|
| 4 3<br>.*..<br>*.**<br>**..<br>****<br>2 2 3 4<br>3 1 3 1<br>1 1 2 2 | 3<br>1<br>2 |